

# Diagnosing Optimization During Fine-Tuning

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## I. SETUP

For this task, DistilBERT was trained on the SST-2 sentiment classification dataset (inspired by Devlin et al. training BERT on SST-2 [1]). The model was fine-tuned in 3 epochs, with a batch size of 32, and seed 42. Two separate models were trained with two different optimizers: AdamW with  $lr=5e-5$  on all parameters, and Muon [2] with  $lr=2e-4$  on the hidden parameters paired with AdamW with  $lr=5e-5$  on the nonhidden parameters. These learning rates were achieved through a sweep in orders of magnitudes of 10.

## II. OPTIMIZATION AND PERFORMANCE METRICS

Quite a few metrics were recorded for a list of reasons:

- 1) **Periodic test loss/acc** This metric was recorded by running the test loop within the train loop every  $n$  runs. Since the model is being fine-tuned, training on too many epochs would lead to overfitting. Hence, evaluating the model at a step frequency is much more reasonable and provides deeper insight to the pattern between epochs.
- 2) **Per step training loss** This is a fundamental optimization trajectory to check for stability and behaviors of training.
- 3) Gradient, update, and weight norms were all captured for the **relative update** as the cross-optimizer step size measure [3]. This metric is used to measure behaviors between optimizers in a more fair manner than just the learning rates.
- 4) **Distance from pre-trained weights** was inspired by Qu et al. to measure the changes that post-training has on the pre-training weights [4].

## III. FINDINGS

TABLE I  
COMPACT SUMMARY TABLE (ON TEST SET)

| Method      | Best Acc. (step)    | Min Loss (step)     | Final Acc.   |
|-------------|---------------------|---------------------|--------------|
| <b>Muon</b> | <b>0.920 (3300)</b> | <b>0.229 (2000)</b> | <b>0.901</b> |
| AdamW       | 0.915 (3900)        | 0.234 (600)         | 0.891        |

**Q1** Muon is ahead on every column by 0.5-1%. However, this setup has only been tested on `seed=42`, and a data-order change alone shifted accuracy by  $\sim 1.5\%$  in a control run, hence the margin is within noise. It is **highly recommended** to refer to `rotational_equilibrium.png`, `periodic_test_eval.png`, and `weight_norms.png` in the `compare/` folder in the repository to differentiate optimizer behavior.

TABLE II

SHARPNESS AND GENERALIZATION GAP AT THE END OF FINE-TUNING. LOSS RISE IS UNDER GAUSSIAN WEIGHT PERTURBATION WITH  $\sigma = 0.05$  ON ALL PARAMETERS; GAP IS TEST MINUS TRAIN CROSS-ENTROPY.

| Method | Top eigenvalue |              | Loss rise      | Train/test   |
|--------|----------------|--------------|----------------|--------------|
|        | hidden         | all          | (all)          | loss gap     |
| AdamW  | <b>10.90</b>   | <b>13.77</b> | 0.00294        | <b>0.240</b> |
| Muon   | 17.25          | 19.79        | <b>0.00173</b> | 0.302        |

**Q2** The top Hessian eigenvalue [5] in Table II measures the steepest direction and finds Muon sharper (17.25 vs. 10.90). From the full sharpness summary table in the repository, the random perturbation [6] measures average curvature surrounding the trained models and finds Muon flatter (paired diff =  $+0.0012 \pm 0.0006$ , 8/10 draws). The interpolation figure (Fig. 1) agrees with the eigenvalue: no barrier between the solutions, AdamW is in a shallower valley, and Muon is in a sharper valley with a cliff on the right.

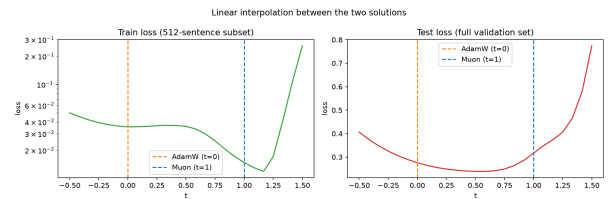


Fig. 1. The linear interpolation between the two trained models in weight space, in addition to covering landscape before and after the models to get a better understanding.

**Q3** There were quite a few limitations with this experiment. First, this experiment was run on one seed due to hardware limitations and time constraints. Then, there was no held-out set during the training (validation set), so the results might be slightly optimistic. The sharpness estimator ran on a 512 subset of the train set. The learning rate sweeps were  $10\times$  spaced, and wasn't done thoroughly. And lastly, from the `weight_norms.png` in the `compare/` folder in the repository, the weight decay of Muon on the hidden parameters acted  $4\times$  harder than AdamW (same weight decay, but  $4\times$  the learning rate), which wasn't accounted for. The conclusions that don't depend on this: Muon's relative update is flat, and a fraction of AdamW's (how the optimizer works); both default pre-training learning rates fail on fine-tuning; the two solutions share a basin; Muon's solution is deeper with a larger generalization gap.

**Q4** The changes to be made for a larger study: (1) more seeds with mean and std; (2) finer optimizer sweeps including aux lr (for Muon) and weight decay; (3) learning rate scheduler; (4) longer runs to test whether rotational equilibrium is reachable in fine-tuning.

## REFERENCES

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